Hien Dang

hiendang@utexas.edu | Google Scholar
Department of Statistics and Data Science, The University of Texas at Austin

Education

University of Texas at Austin 2024 – Present
Ph.D. in Statistics, Aug 2024 – May 2029 (expected)

- Advisors: Alessandro Rinaldo and Pratik Patil
- GPA: 4.0/4.0

University of Science - Ho Chi Minh City, Vietnam 2016 – 2020
Bachelor of Science in Mathematics, Honors Program

Research Interests

- High-dimensional statistics, statistical machine learning, random matrix theory, spectral algorithms, cross-validation and tuning.
- Statistical foundation for knowledge distillation and learning with model-generated data, neural scaling laws and other phenomena in deep learning (e.g., neural collapse, posterior collapse).

Preprints and Ongoing work

1. **Hien Dang**, Pratik Patil, Alessandro Rinaldo. "Minimax Rates and Spectral Instance Optimality under Perturbed Power-Law Covariance", *Work in progress, arXiv preprint expected Oct 2026.*
2. Long Tung Vuong, Vy Vo, **Hien Dang**, Van-Anh Nguyen, Thanh-Toan Do, Mehrtash Harandi, Trung Le, Dinh Phung. "Why Domain Generalization Fails? A View of Necessity and Sufficiency". *arXiv preprint (2025).* arXiv

Publications

(*) denotes equal contribution

1. **Hien Dang**, Pratik Patil, Alessandro Rinaldo. "Prediction-Only Distillation in Linear and Logistic Regression". *Conference on Neural Information Processing Systems (NeurIPS), 2026.* arXiv
2. **Hien Dang**, Pratik Patil, Alessandro Rinaldo. "Optimal Unconstrained Self-Distillation in Ridge Regression: Strict Improvements, Precise Asymptotics, and One-Shot Tuning", *International Conference on Machine Learning (ICML), 2026.* arXiv
3. Long Tung Vuong, Julien Monteil, **Hien Dang**, Volodymyr Vaskovych, Trung Le, Vu Nguyen. "On the Mechanisms of Collaborative Learning in VAE Recommenders", *International Conference on Learning Representations (ICLR), 2026.* arXiv
4. **Hien Dang**, Tho Tran, Tan Nguyen, Nhat Ho. "Neural Collapse for Cross-entropy Class-Imbalanced Learning with Unconstrained ReLU Features Model", *International Conference on Machine Learning (ICML), 2024.* arXiv
5. **Hien Dang**, Tho Tran, Tan Nguyen, Nhat Ho. "Beyond Vanilla Variational Autoencoders: Detecting Posterior Collapse in Conditional and Hierarchical Variational Autoencoders", *International Conference on Learning Representations (ICLR), 2024.* arXiv
6. **Hien Dang***, Tho Tran*, Stanley Osher, Hung Tran-The, Nhat Ho, Tan Nguyen. "Neural Collapse in Deep Linear Networks: From Balanced to Imbalanced Data", *International Conference on Machine Learning (ICML), 2023.* arXiv

Service

- **Journal Reviewer**: Annals of Statistics (AoS); SIAM Journal on Mathematics of Data Science (SIMODS).
- **Conference Reviewer**: International Conference on Machine Learning (ICML); Conference on Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR); International Conference on Artificial Intelligence and Statistics (AISTATS).

Teaching Experience

Teaching Assistant, UT Austin

- SDS 336: Practical Machine Learning (undergraduate) — Fall 2026
- SDS 322E: Elements of Data Science (undergraduate) — Spring 2026
- SDS 391P.3: Theory of the Linear Model (graduate) — Fall 2025
- SDS 391P.1: Advanced Statistical Modeling and Applications I (graduate) — Fall 2025
- DSC 382: Foundations of Regression and Predictive Modeling (graduate) — Spring 2025
- SDS 324E: Elements of Regression Analysis (undergraduate) — Fall 2024

Other Work Experience

- AI Research Resident, FPT Software AI Center - Ho Chi Minh City, Vietnam — 2022 – 2024
- Actuarial Analyst, Sun Life Financial - Ho Chi Minh City, Vietnam — 2019 – 2022
- Actuarial Candidate, Society of Actuaries - Record — 2019 – 2022

Award and Scholarship

Highest GPA, 2016–2020 mathematics cohort at University of Science, Ho Chi Minh City — 2020
Vietnam Institute for Advanced Study in Mathematics (VIASM) Scholarship — 2018, 2019, 2020
Silver Medals, Vietnam Mathematics Olympiad for Undergraduate Students — 2017, 2018
Odon Vallet Scholarship — 2017
Ranked 1st among 7,400 candidates in Soc Trang City, National University Admission Exam — 2016